MAHATMA GANDHI UNIVERSITY, KOTTAYAM FIRST SEMESTER
MGU-UGP (HONORS) REGULAR
EXAMINATION NOVEMBER 2024
First Semester
Discipline Specific Core Course - MG1DSCSDV100 - FUNDAMENTALS OF ACOUSTICS (2024 ADMISSION ONWARD) ANSWER KEY

Part A: Very Short Answer Questions

Answer any ten questions. Each question carries 2 marks.

1. **What is a transverse wave? [U] [1]**

A wave in which particles of the medium move perpendicular to the direction of wave propagation.

2. **What is sound? [U] [1]**

Sound is a mechanical wave that propagates through a medium due to the vibration of particles.

3. **The device used to measure the sound pressure level? [U] [1]**

Sound Level Meter/Spl Meter.

4. **Why is there no sound in space? [U] [1]**

Space is a vacuum, and sound requires a medium (like air or water) to propagate.

5. **Factors affecting the speed of sound in a medium? [An] [1]**

Medium's density, temperature, elasticity, and humidity.

6. **What is the frequency of a sound wave? [U] [1]**

The number of vibrations or cycles per second, measured in Hertz (Hz).

7. **Define a harmonic sound. [U] [2]**

A sound that contains frequencies that are integer multiples of a fundamental frequency.

8. **Define timbre. [U] [2]**

The quality of a sound that distinguishes it from other sounds with the same pitch and volume.

9. **What is an echo? [K] [4]**

The reflection of sound when it bounces back after hitting a surface and is heard distinctly.

10. **What is the refraction of sound? [K] [4]**

The bending of sound waves as they pass through different mediums with varying densities.

11. **What is diffraction? [K] [4]**

The bending and spreading of sound waves around obstacles or through openings. 12. **What is a critical band? [K] [6]**

The critical band is the smallest range of frequencies that activate the same part of the basilar membrane.

Part B: Short Answer Questions

Answer any six questions. Each question carries 5 marks.

13. **Explain the source, medium, and receiver of sound.** [U] [1]

Source: Object generating vibration.
Medium: Substance (air, water, etc.) through which sound travels.
Receiver: Device or ear that detects the sound.

14. **What is pitch? How is frequency related to pitch?** [An] [1]

Pitch: The perceived highness or lowness of sound.
Higher frequency = higher pitch; lower frequency = lower pitch.

15. **Define medium density and its importance in sound transmission.** [U] [3]

Medium density: Mass per unit volume of the medium.
Higher density slows sound; lower density allows faster propagation.

16. **What is a wave equation?** [U] [1]

Wave equation in terms of velocity: $v = f\lambda$, where v is velocity, f is frequency, and λ is wavelength.
Another form of the wave equation:

$$\frac{\partial^2 p}{\partial t^2} = c^2 \nabla^2 p$$

Where:

- p is the acoustic pressure.
- t is time.
- c is the speed of sound in the medium.

17. **Describe the role of harmonics in determining timbre.** [A] [2]

Harmonics add richness, distinctiveness, and tone quality. Different instruments produce different harmonic structures.

18. **What is the law of reflection?** [K] [4]

Angle of incidence = angle of reflection. Sound follows the same principles as light.

19. **Describe an example of diffraction in everyday life.** [A] [4]

Hearing a train around a corner due to sound bending. Sound waves spreading through a doorway.

20. **How can we soundproof a room?** [A] [5]

Use dense materials, acoustic panels, carpets, and double-glazed windows. Seal gaps and cracks.

21. **What is the function of the Eustachian tube?** [U] [6]

Balances pressure between the middle ear and the external environment. Ensures proper vibration of the eardrum.

Part C: Essay Questions

Answer any two questions. Each question carries 10 marks.

22. **Explain the differences between simple and complex sounds. How do harmonics and partials influence the richness of complex sounds? [An] [2]**

Simple sound: Single frequency (pure tone).

Complex sound: Multiple frequencies (harmonics and partials).

Harmonics create smooth sounds (musical instruments).

Partials (inharmonic) add roughness (noise, bells).

23. **Describe the stages of a sound envelope and explain their influence on musical sound perception. [E] [2]**

Attack: Time to reach maximum intensity.

Decay: Reduction after the peak.

Sustain: Constant level duration.

Release: Time taken to fade after stopping.

Examples: Piano (fast attack) vs. violin (gradual attack).

24. **Discuss the fundamental characteristics of a sound wave, including amplitude, frequency, wavelength, and velocity. [An] [1]**

Amplitude: Loudness (greater amplitude = louder sound).

Frequency: Pitch (higher frequency = higher pitch).

Wavelength: Distance between wave crests (related to pitch).

Velocity: Speed of sound in the medium.

Example: Sound in air at $20°C$ travels at $343\,\mathrm{m/s}$.